

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Brain Route AI

also called LEARN MATE

Domain of Project – AI, Agentic AI, Education Technology (EdTech)

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Problem Statement -

Problem Statement 12 :Agentic AI for Personalized Course Pathways

The Challenge

Students often struggle to identify the right learning path that aligns with their interests and long-term goals due to the overwhelming number of online courses and a lack of personalized guidance. **Brain Route AI** aims to solve this by acting as an Agentic AI coach that interacts with students, understands their interests (like Frontend Development, Cybersecurity, UI/UX Design, etc.), assesses their current skill level, and dynamically builds a personalized course roadmap that adapts over time based on progress and preferences

The Objective

Build an AI-powered personalized course pathway assistant called **Brain Route AI** using IBM Cloud, Prompt Lab, IBM watsonx Orchestrate ,and IBM Granite Models.

The system should:

- Recommend personalized learning pathways based on student interests and career goals
- Analyze skill levels and identify missing competencies
- Suggest relevant courses, certifications, and learning resources
- Adapt learning roadmaps based on progress and preferences
- Provide career-focused recommendations for industry readiness
- Help students stay motivated with progress tracking and milestone visualization

Technology Used

Tech Stack (Mandatory) Use of IBM Cloud Lite services / IBM Granite is mandatory

I used:

1. IBM Cloud lite services
2. IBM watsonx Orchestrate
3. Prompt Lab
4. IBM Granite Models
5. IBM watsonx.ai Runtime

IBM Cloud + Prompt Lab + IBM Granite Models

IBM CLOUD + IBM watsonx Orchestrate +IBM watsonx.ai Studio +IBM watsonx.ai Runtime

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent

Brain Route AI is an intelligent Agentic AI-based learning assistant designed to help students discover the right educational and career pathway according to their interests and goals.

The system uses IBM Granite models through Prompt Lab to understand student preferences, analyze skill levels, and generate personalized learning roadmaps.

The solution works through multiple AI agents collaborating together to deliver adaptive and intelligent learning guidance. The system can recommend courses, certifications, projects, and skill-building activities based on domains such as Frontend Development, UI/UX Design, Cybersecurity, Data Science, AI/ML, and Cloud Computing.

The platform continuously adapts recommendations based on user progress, completed courses, and changing interests. This creates a dynamic and personalized learning experience similar to having a real AI mentor.

The solution also integrates RAG-based retrieval to fetch accurate and relevant course information from trusted learning platforms and educational resources

Overall, Brain Route AI acts as a smart digital career mentor that helps students make informed learning decisions and achieve their long-term goals effectively.

Technology Used

IBM Cloud

Provides a secure and scalable cloud environment for running AI-powered educational solutions and deploying intelligent services.

Watsonx Orchestrate

Used for Agentic AI workflow Conversational interaction AI orchestration User guidance

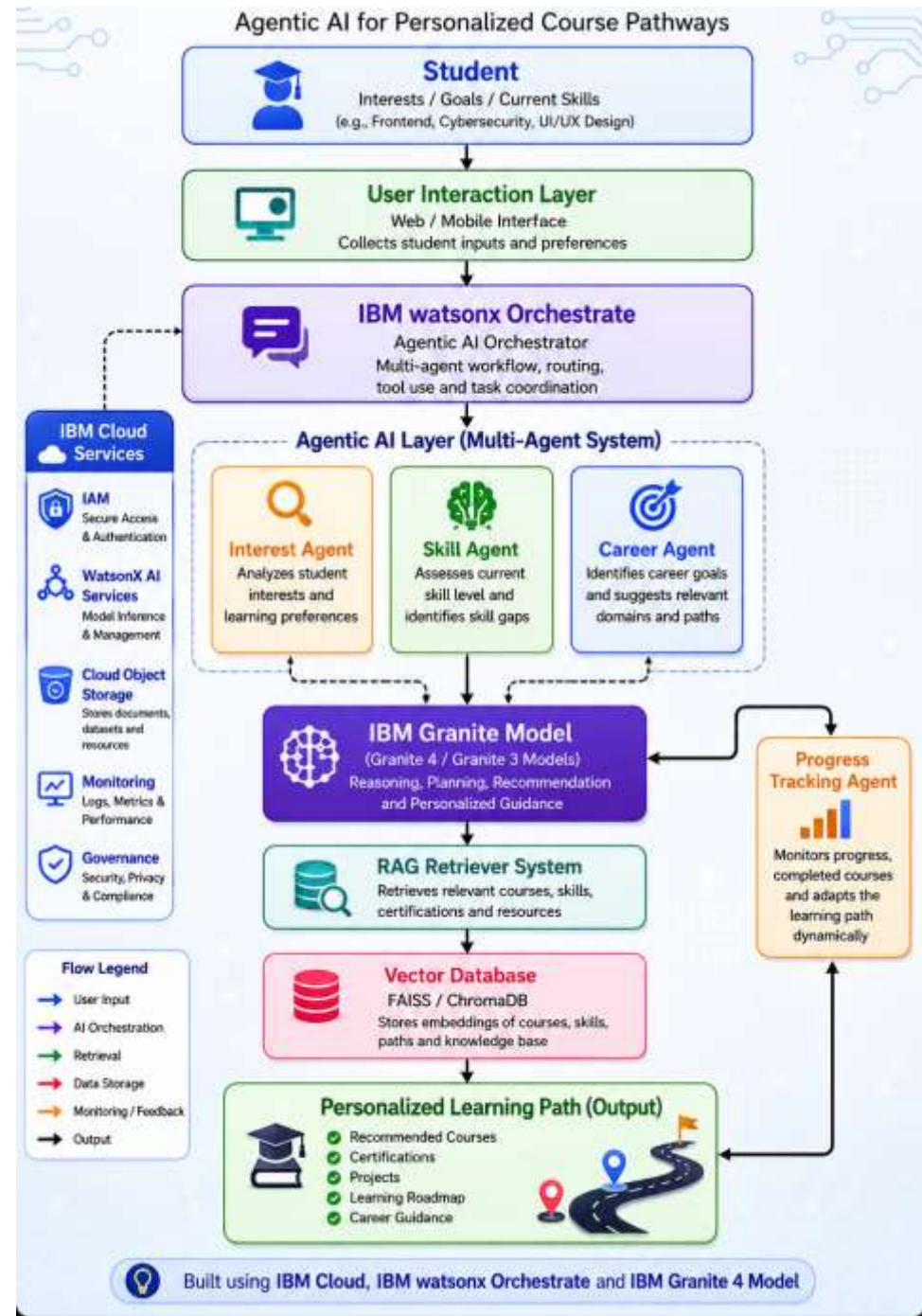
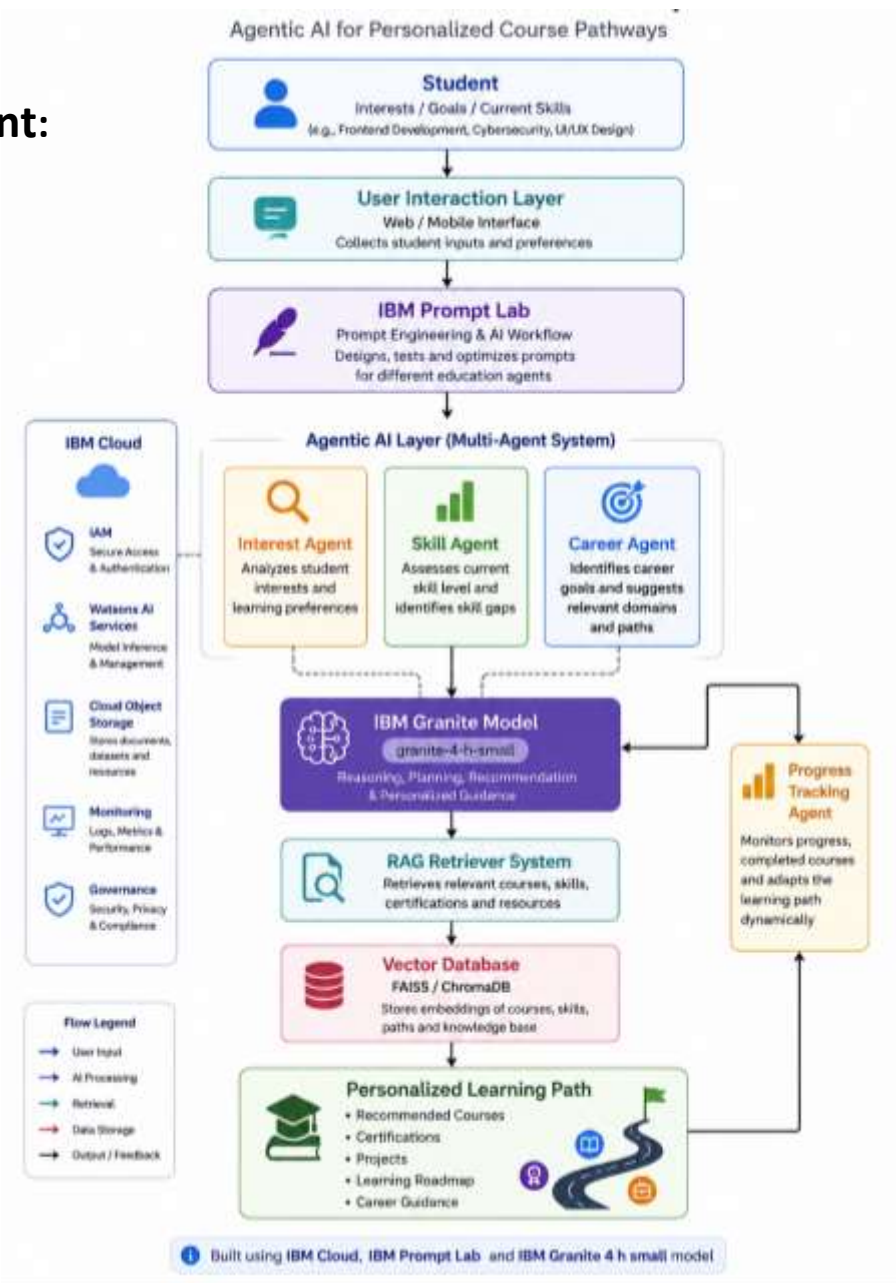
IBM Prompt Lab

Used for prompt engineering, testing prompts, refining AI responses, and generating personalized learning recommendations.

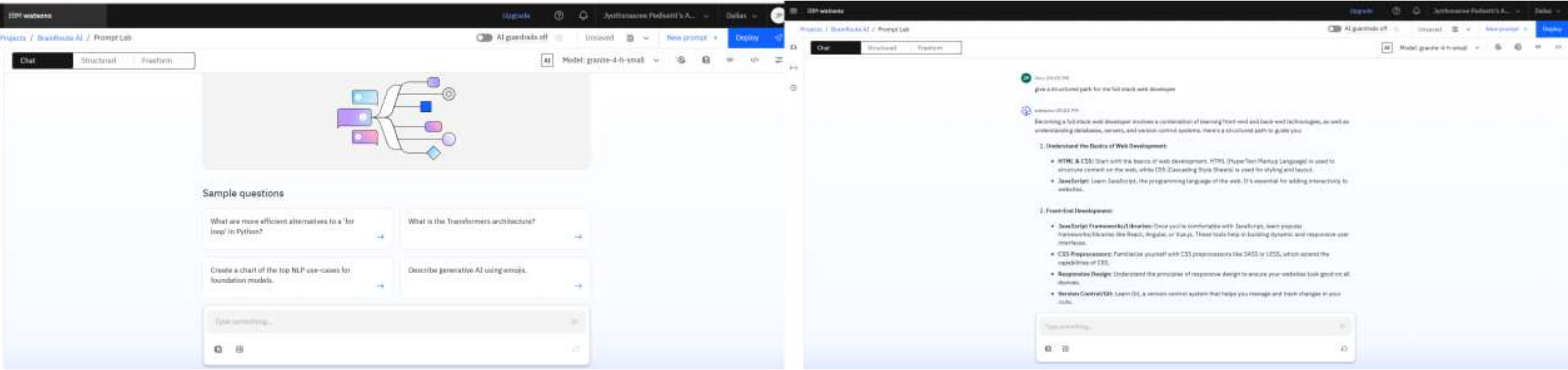
IBM Granite Models

Used for reasoning, roadmap generation, personalized recommendations, summarization, and intelligent conversational guidance.

Architecture Blueprint:



WORKING SCREENSHOTS: Prompt lab and watsonx Orchestrate



The screenshot shows the IBM Cloud Resource list page. It features a table with columns for Name, Group, Location, Product, Status, and Tags. The table lists two resources under the 'AI / Machine Learning' group: 'watsonx.ai Runtime-yii' and 'watsonx.ai Studio-ng'. Both resources are located in 'Dallas (us-south)' and are in an 'Active' status.

Name	Group	Location	Product	Status	Tags
watsonx.ai Runtime-yii	Default	Dallas (us-south)	watsonx.ai Runtime	Active	1
watsonx.ai Studio-ng	Default	Dallas (us-south)	watsonx.ai Studio	Active	1

The screenshot shows the BrainRoute AI interface, which is an AI Personalized Learning Roadmap Generator. It features a form with three input fields: 'Student Interest', 'Select Skill Level', and 'Career Goal'. A blue button labeled 'Generate Roadmap' is positioned below the form. The interface is dark-themed with white text and a blue button.

IBM watsonx Orchestrate

BrainRoute AI

AI-powered personalized learning assistant developed using IBM watsonx Orchestrate and IBM Granite models. It generates personalized career pathways, skill roadmaps, course recommendations, and project guidance for students.

Open Live AI Demo

Profile

Knowledge

Tutorial

Behavior

Channels

Profile

Learn the what your agent does and how to use it.

Recommend

Welcome message

History message

Quick start prompts

Recommended courses based on my skill level.

Build with AI

Talk to agent

Running in debug mode

Today

You 6:23 PM

Help me choose between AI/ML and Cybersecurity.

Brain Route AI 6:23 PM

Let's Figure Out What's Right for You

Below are a few quick questions that will help us pinpoint which path—AI/ML or Cybersecurity—fits your interests, goals, and current experience best. Feel free to answer as much or as little as you like; even a couple of answers will give us a solid starting point.

Question	Why It Matters
What excites you more? - Building models that <i>predict, classify, or generate</i> data (e.g., chatbots, recommendation engines). - Protecting systems, finding vulnerabilities, and defending against attacks.	Shows your natural curiosity and motivation.
How do you feel about math & statistics? - Comfortable with linear algebra, calculus, probability, and statistics. - Prefer logical reasoning, networking concepts, and system architecture.	AI/ML relies heavily on quantitative modeling; Cybersecurity leans on logical problem-solving and architecture.
Do you enjoy coding for data or for security? - Data-heavy Python/R, TensorFlow/PyTorch, Jupyter notebooks. - Scripting for scanning, automation, incident response (Python, Bash, PowerShell).	Determines which tool-chains feel more natural.
What kind of impact do you want to make? - Enable smarter products, automate decisions, create new AI services. - Safeguard privacy, protect critical infrastructure, stop cyber-crime.	Aligns career purpose with domain outcomes.
How do you see your future work environment? - Research labs, data-driven product teams, start-ups building AI products. - Security operations centers (SOC), consulting firms, compliance	Helps match typical job settings.

Type something...

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

14 days left

AI

Agents

Brain Route AI

Chats

Today

Recommend courses based on my sk...

Create new agent

Manage agents

Agent memory

Preview

Today

You 6:32 PM

Recommend courses based on my skill level.

Brain Route AI 6:32 PM

Sure! I'd love to help you map out the perfect learning plan. To recommend courses that match **your current skill level** and **career goals**, could you tell me a bit more about yourself?

1. What areas of technology interest you the most?

In g., Front-end Development, Back-end Development, UI/UX Design, Cybersecurity, AI/ML, Data Science, Cloud Computing, Mobile App Development

2. How would you describe your experience right now?

• Beginner – just starting, little to no hands-on experience

• Intermediate – comfortable with basics, have built a few projects or taken an introductory course

• Advanced – strong foundation, several projects or work experience, looking to specialize or deepen expertise

Type something...

Role of Agentic AI in the solution

Role of Agentic AI in this BRAIN ROUTE AI :

Agentic AI enables Brain Route AI to behave like an intelligent autonomous career mentor instead of a basic chatbot.

The system uses multiple AI agents that collaborate together to deliver personalized educational guidance. Each agent performs a specialized task such as understanding user interests, recommending courses, generating roadmaps, and tracking progress.

The AI system continuously adapts based on user interactions and learning progress, making recommendations more accurate and personalized over time.

Agentic AI also helps in decision-making by analyzing multiple factors including skill levels, learning pace, career goals, and industry trends.

This creates a proactive, adaptive, and goal-driven learning ecosystem that helps students stay focused and career-ready.



Technology Used

- **IBM Granite model** - IBM-granite 4-h small is used For natural language understanding, reasoning, and personalization
- **IBM watsonx Orchestrate**
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Brain Route AI efficiently. IBM Cloud enables a reliable, secure, and scalable environment for deploying the Agent.
- **RAG**– RAG to fetch real data first, then generate intelligent answers, making the AGENTIC AI COUCH more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance

Novelty and Uniqueness

Personalized Learning Roadmaps

Generates adaptive course pathways according to each student's goals and skill levels.

Dynamic Recommendation System

Continuously updates recommendations based on progress and preferences.

RAG-Based Retrieval

Combines retrieval with AI generation to provide accurate and trustworthy learning suggestions.

Career-Focused Guidance

Aligns learning recommendations with industry-required skills and certifications.

Scalable and Intelligent

Built using IBM Cloud and IBM Granite Models for enterprise-grade reliability and scalability.

AI Mentor Experience

Acts like a personalized AI learning coach available anytime for guidance and support

Git Hub Link

<https://github.com/jyothsna0012/Brain-Route-AI->

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Future Scope

1. Integration with Job Platforms

The system can connect with LinkedIn, Internshala, and other job platforms to recommend skills according to current industry demands.

2. AI Interview Preparation

Can provide mock interviews, resume analysis, and career preparation guidance using AI.

3. Multilingual Learning Support

- Support for regional languages to make learning accessible to more students.

4. Integration with Learning Platforms

- Can integrate with Coursera, Udemy, edX, and YouTube Learning resources for automated recommendations.

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Thank You!

Thank you for your time and interest.